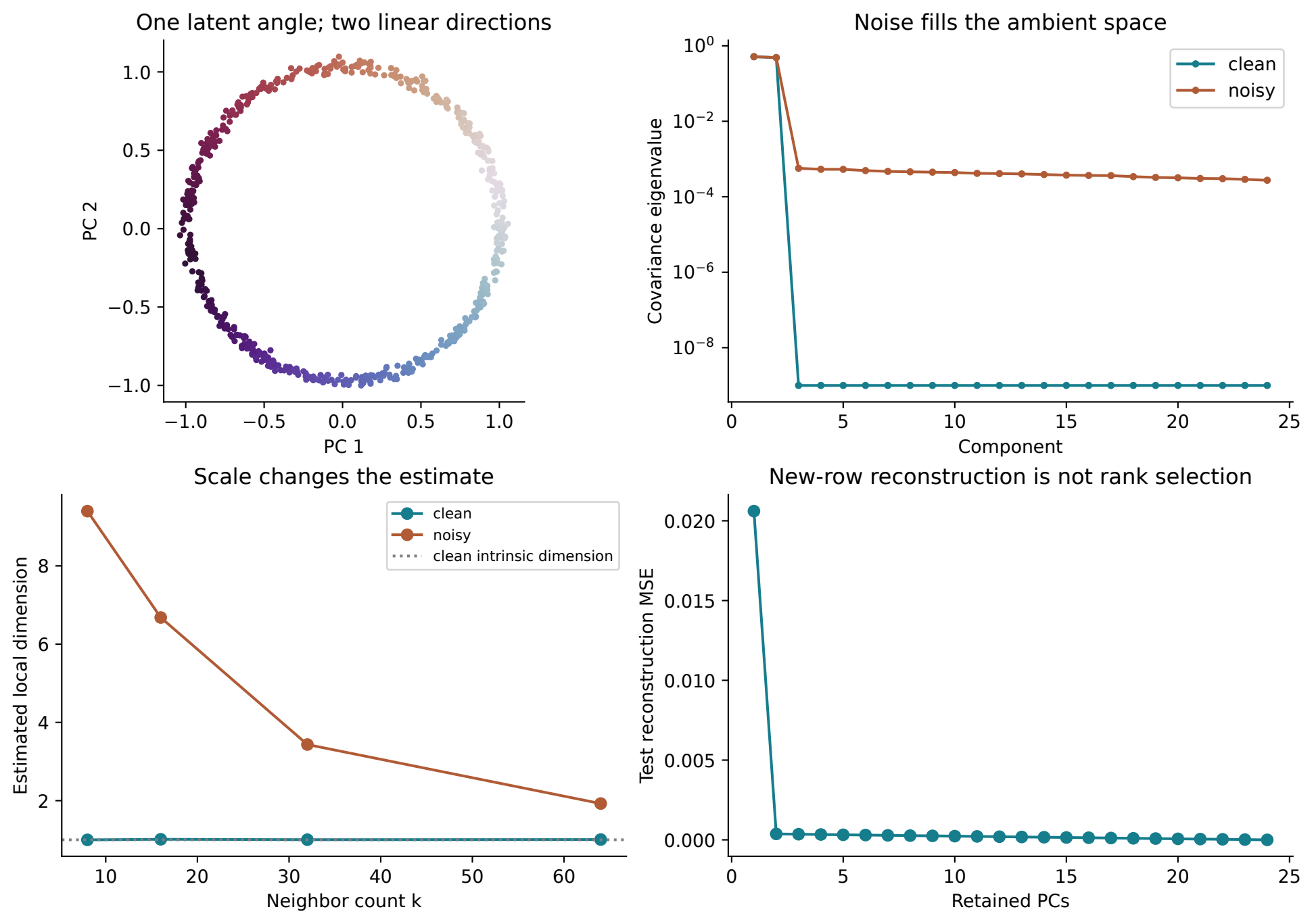
One latent angle; two linear directions
PC 2
PC 1
1.0
0.5
0.0
−0.5
−1.0
Noise fills the ambient space
Covariance eigenvalue
Component
clean
noisy
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
5
10
15
20
25
Scale changes the estimate
Estimated local dimension
Neighbor count k
clean
noisy
clean intrinsic dimension
2
4
6
8
10
20
30
40
50
60
New-row reconstruction is not rank selection
Test reconstruction MSE
Retained PCs
0.000
0.005
0.010
0.015
0.020